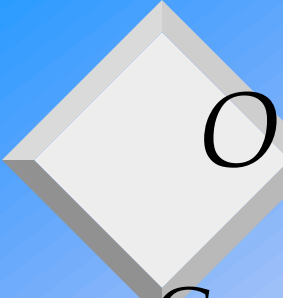




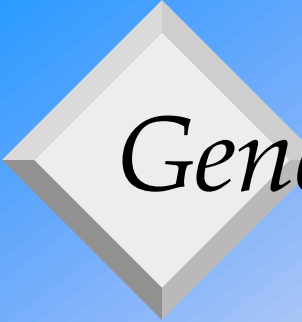
Data Warehouse Architectures





Objectives of Today's Lecture

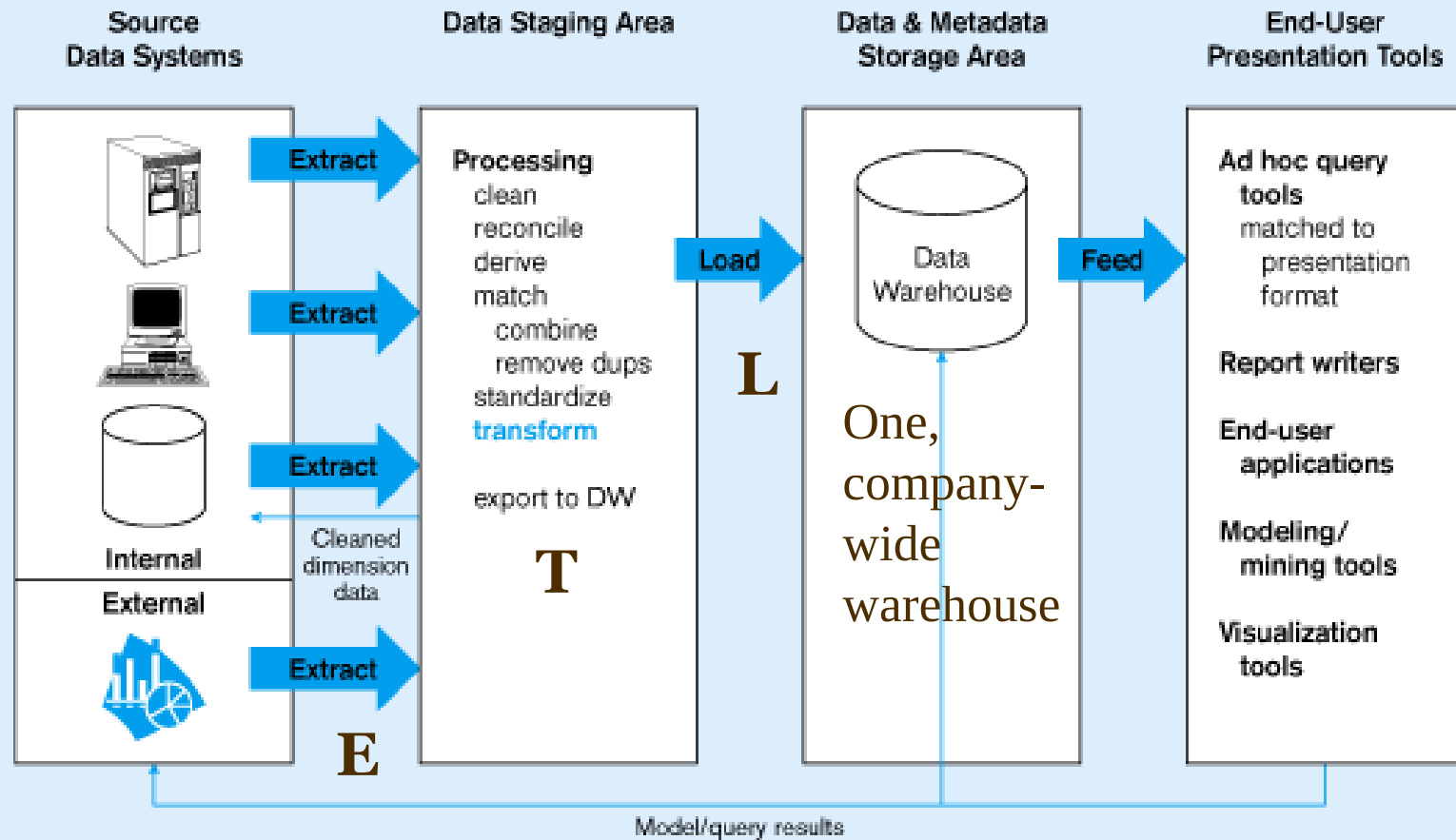
- Generic Two-Level Architecture
- Independent Data Mart
- Dependent Data Mart and Operational Data Store
- Logical Data Mart and @ctive Warehouse
- Three-Layer architecture



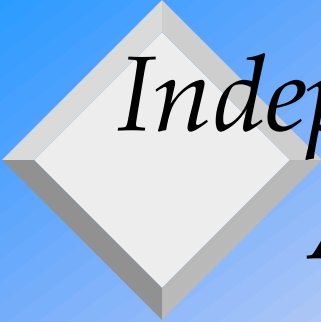
Generic Two-level Architecture

- This architecture has following basic functions
 - Data extraction form various internal and external operational systems
 - Transformation and integration of data before loading.
 - Transactions may be sent to source systems to correct errors discovered in data staging.
 - Contains both detailed and summary data
 - Results after the query may be fed back to DWH and operational DBs
- Extraction and loading happens on periodic basis

Generic two-level architecture



Periodic extraction → data is not completely current in warehouse



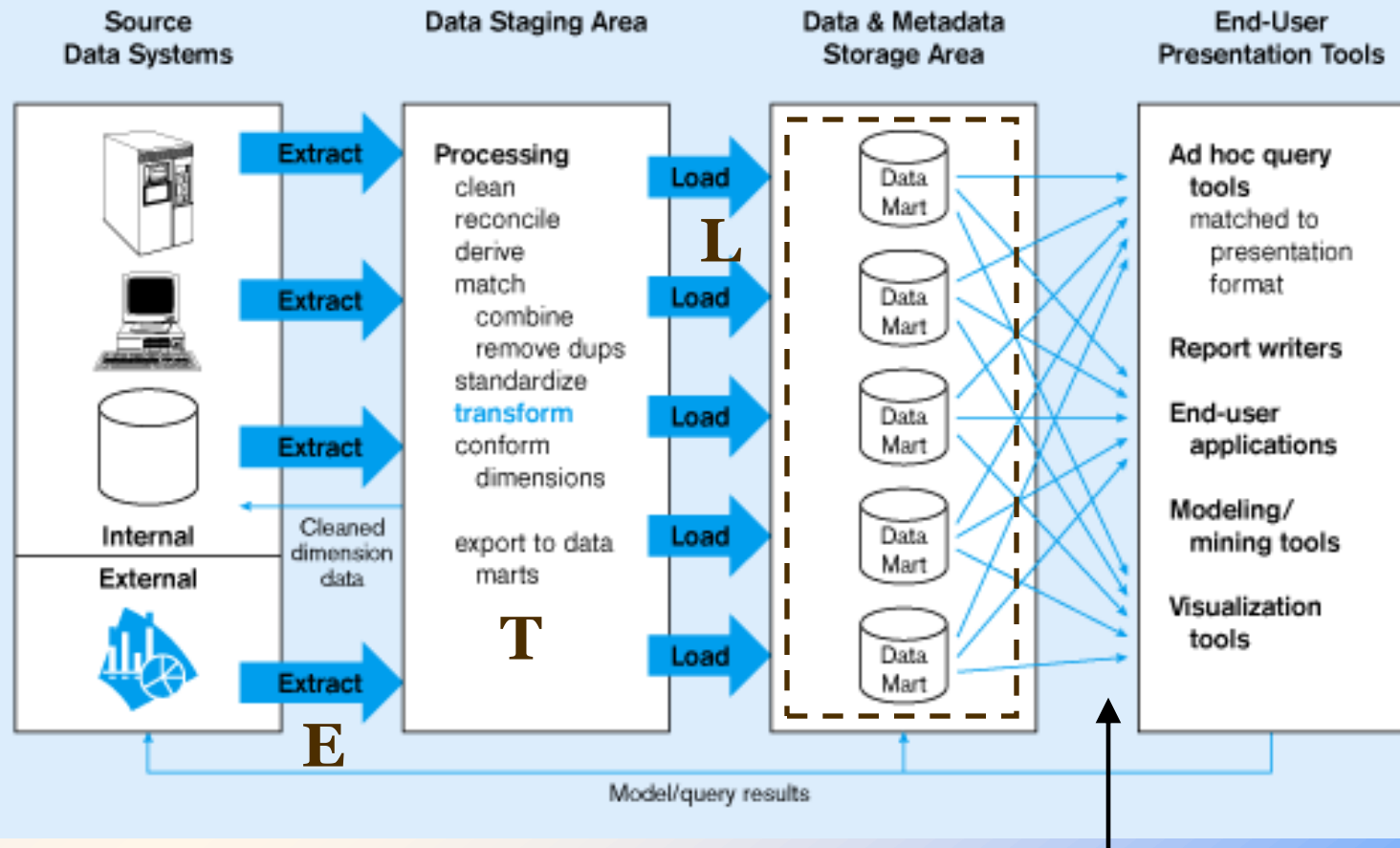
Independent Data Mart DWH Architecture

- What is Data mart ?
 - E.g. of data mart
 - Marketing data mart
 - Finance data mart
 - Supply chain data mart
- Independent Data mart
 - A data mart filled with data extracted form operational environment, without benefit of data warehouse
- In this architecture
 - The complexity for users is reduced
 - Complexity of ETL is increased
 - Scalability issue due to technology constraint
 - Separate ETL for each data mart
 - Data marts are not consistent with each other

Independent Data Mart architecture

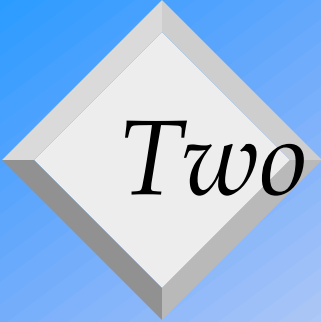
Data marts:

Mini-warehouses, limited in scope



Separate ETL for each
independent data mart

Data access complexity
due to *multiple* data marts



Two Debates w.r.to Architecture

- Phased approach towards implementing DWH. Essence of this debate is whether each data mart should or should not evolve as subset of enterprise-wide definition of decision support system.
- Deals with the suitable DB architecture for analytical processing. It concerns on the extent to which a data mart DB should be normalized.



Dependent Data Mart and Operational Data Store Architecture

- Dependent data mart
 - A data mart filled exclusively from enterprise DWH and its reconciled data
- Enterprise DWH
 - A centralized, integrated DWH that is the control point and single source of all data made available to end users for decision support applications



Operational Data Store (ODS)

- Definition
 - An integrated, subject oriented, updatable, current valued, detailed DB designed to serve operational users as they do decision support processing
- It is typically normalized relational DB but tuned for decision making applications
- ODS does not contain history
- ODS is not designed for high volume transaction processing, but rather for decision support
- It also serves the role of data staging area for loading data into EDWH
- It may receive data immediately or with some delay from system.

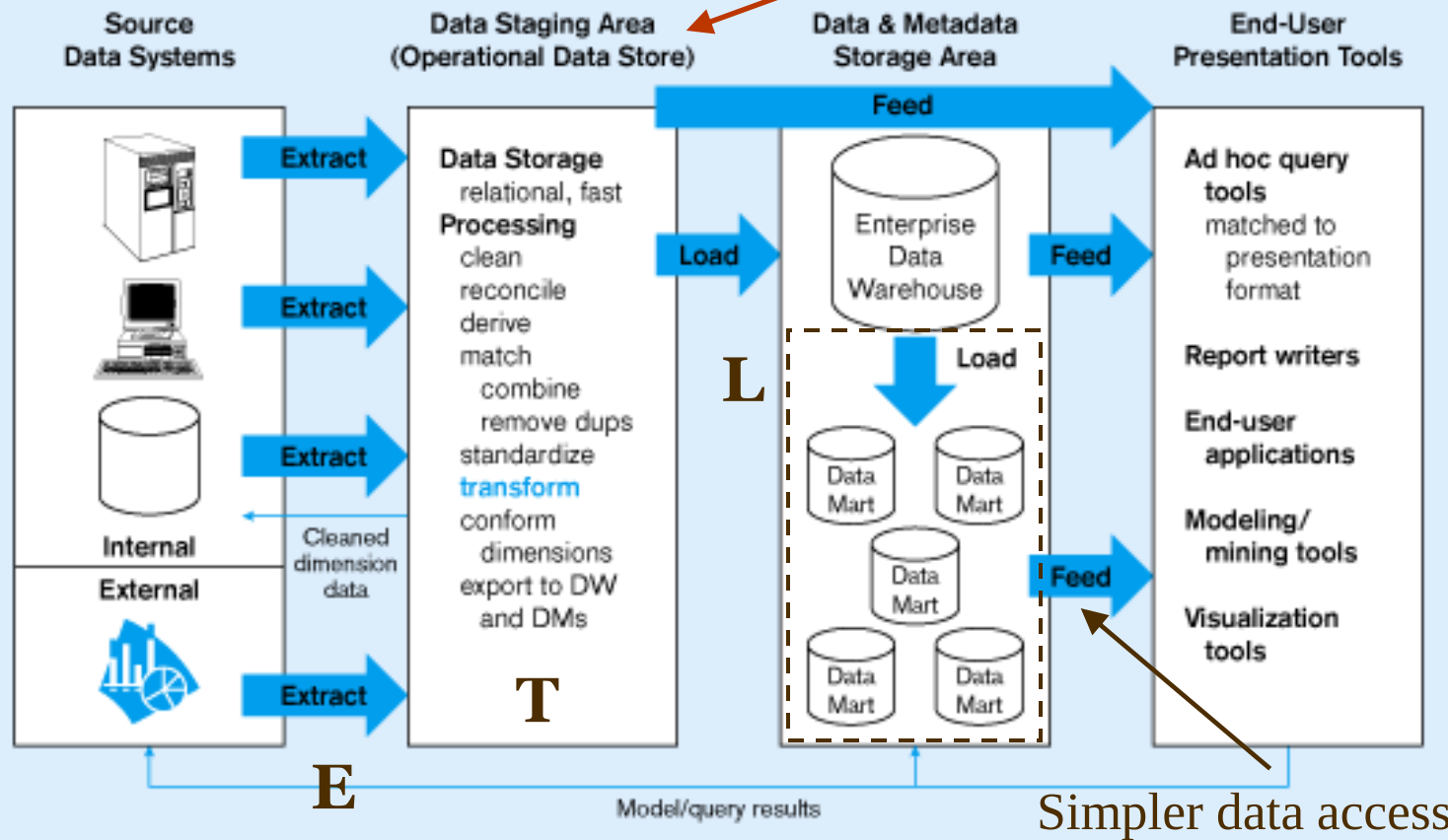


Dependent Data Mart and Operational Data Store Architecture Contd...

- Dependent Data Mart and Operational Data Store Architecture
 - Is also called a corporate information factory (CIF)
 - Has eliminated
 - Redundancy across the data marts
 - Extra work for many ETL processes
 - Technology constraints
 - High cost to keep data marts synchronized

Dependent data mart with operational data store

ODS provides option for obtaining **current** data



Single ETL for
enterprise data warehouse
(EDW)

Dependent data marts
loaded from EDW



Logical Data Mart and @Ctive Warehouse Architecture

- Logical data marts
 - A data mart created by the relational view of a data warehouse
 - They are not physically separate data bases
 - They are different relational views of one physical, slightly renormalized relational DWH
- Data are moved into the DWH rather than a separate staging area to enhance the cleansing and transformational steps
- New data marts can be made very quickly because
 - No physical DB or DB technology is needed
 - No need to write loading routine
-

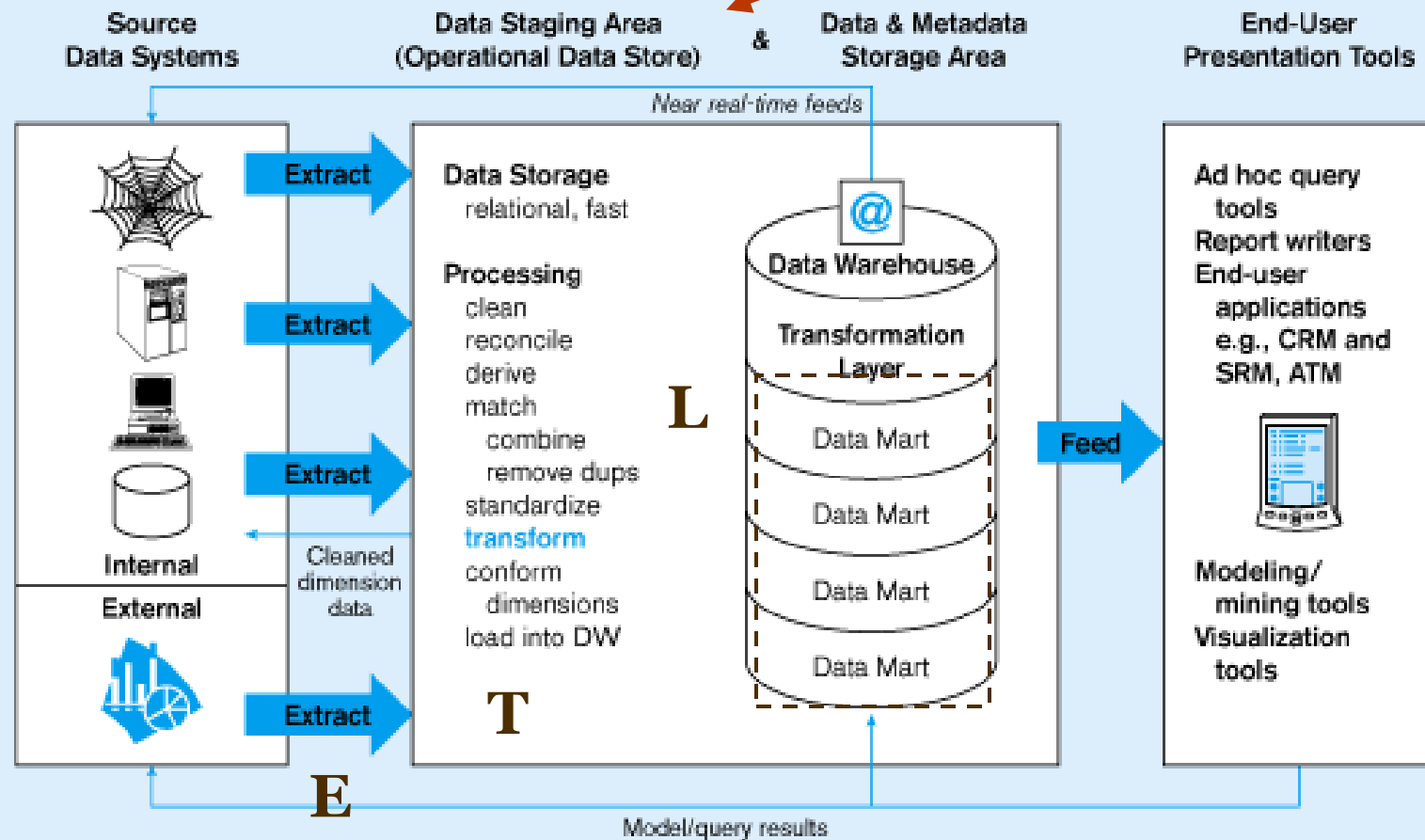


Logical Data Mart and @Ctive Warehouse architecture contd..

- In @ctive DWH
 - The source data systems and data warehouse exchange data at near-real-time pace.
- Some uses of @data warehouse
 - Just in time transportation of rerouting deliveries based on up-to-date inventory levels
 - Fraud detection in credit cards
 - E.g. unusual pattern of transactions could alert a sales clerk or on-line shopping cart routine to take extra precautions
- It is easier to drill down because ODS and EDWH are the same

Logical data mart and @ctive data warehouse

ODS and data warehouse
are one and the same



Near real-time ETL for
@active Data Warehouse

Data marts are NOT separate databases,
but logical **views** of the data warehouse
➔ Easier to create new data marts



Three-Layer Data Architecture

- Three important terms
 1. Operational data
 2. Reconciled data
 3. Derived data



Three-Layer Data Architecture Contd..

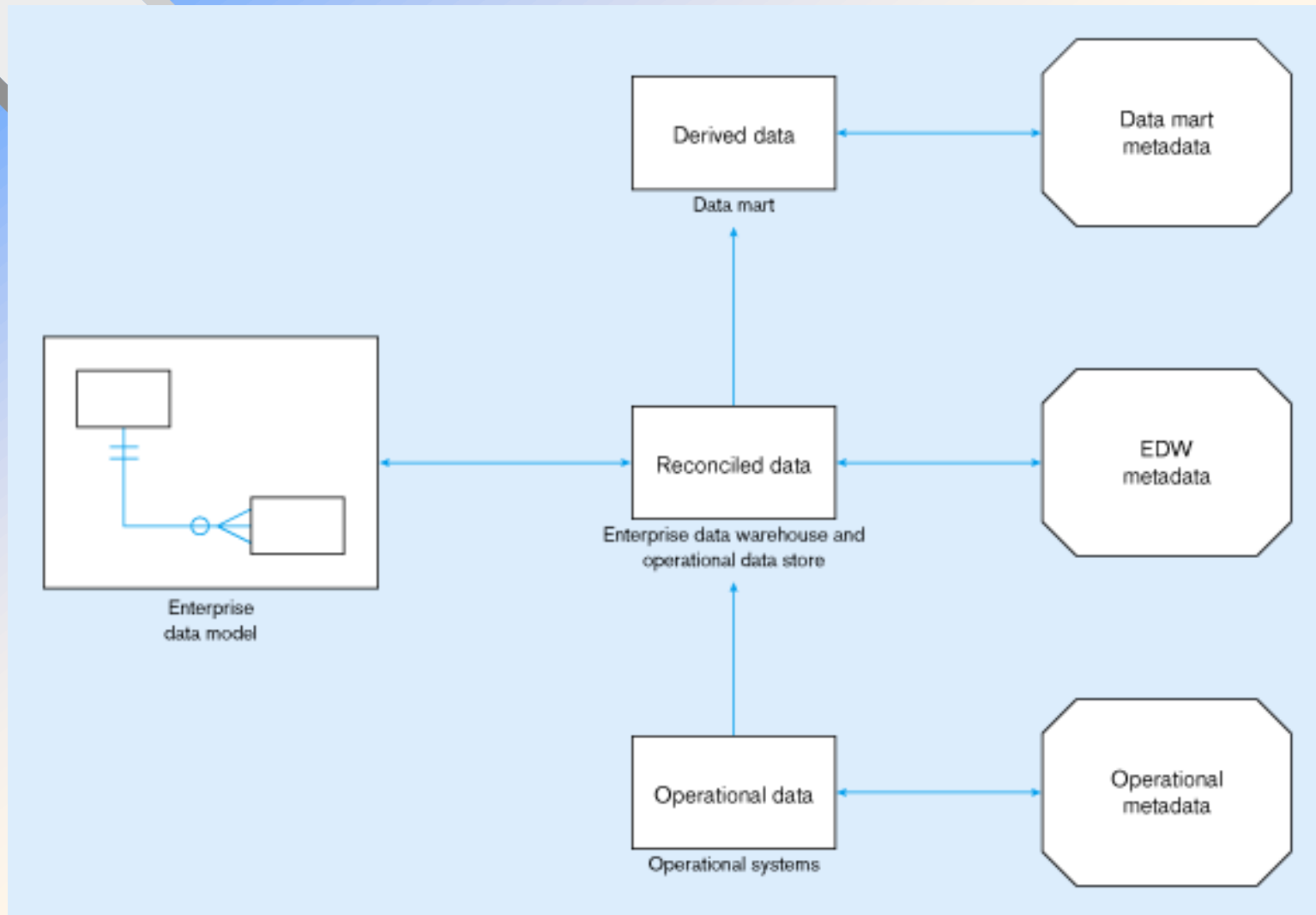
- Operational data ?
- Reconciled data
 - Is detailed, current data intended to be the single, authoritative source for all decision support applications
- Derived data
 - Data that have been selected, formatted, and aggregated for end user decision support application



Three-Layer Data Architecture Contd..

- Two components plays important role in this architecture :
 - Enterprise data model
 - Meta data

Three-layer architecture





Role of Enterprise Data Model

- It presents total picture explaining the data required by an organization
- It control the phased evolution of DWH
- It takes too long to develop the enterprise data model in one step and dynamic need for decision making will change before the warehouse is built



Role Of Meta Data

- Meta data are data that describes the properties or or characteristics of other data
- Operational meta data
 - Describes data in various operational systems
- EDWH meta data
 - Derived from EDWH
 - Describes the reconciled data layer as well as the rules rules for transforming operational data into reconciled data
- Data mart Meta Data
 - Describes the derived data layer and the rules for transforming reconciled data to derived data



Thank You Very Much